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Author(s): William G. Gale

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Economic Effects of Federal Credit Programs

By WILLIAM G. GALE *

Since 1980, the federal government has directly subsidized one-third of all nonfederal borrowing. This paper presents numerical estimates of the effects of federal lending. Existing credit subsidies appear to have important effects on the allocation of credit, but little effect on aggregate investment. Efficiency costs are shown to be large (approximately 1/3 percent of GNP). Government costs exceed 50 cents per dollar of incremental targeted lending. Interactions among programs can eliminate much or all of the original gain provided by a subsidy, especially if borrowers are rationed. The paper also examines the effects of several policy reforms. (*JEL* 313, 321)

From 1980 to 1987, the Federal government borrowed $1,253 billion to cover budgetary shortfalls.[1] Large and persistent fiscal deficits have generated widespread attention in academic and policy debates. Over the same period, federal lending programs extended $1,208 billion of net credit, while new federal and federally assisted lending exceeded $4 trillion. Since 1980, the federal government has subsidized, guaranteed, or directly extended more than one-third of all borrowing by nonfederal sectors. However, despite the potential importance of federal credit activity, there exist few systematic analyses of the effects and appropriate role of the government as a lender. Barry Bosworth et al. (1987) provide a comprehensive discussion of issues and characteristics of lending programs but provide little formal analysis. Gregory Mankiw (1986), David de Meza and David Webb (1987, 1988), Gale (1989, 1990b), and Bruce Smith and Michael Stutzer, (1989) provide theoretical models of the effects of credit subsidies in markets with imperfect information.[2] Charles Calomiris et al. (1986) analyze the empirical effects of agricultural credit policy.

The purpose of this paper is to develop a framework in which to analyze the quantitative effects of federal lending on credit allocation, economic efficiency, and related issues. The underlying model is based on Joseph Stiglitz and Andrew Weiss (1981) and posits asymmetric information between borrowers and lenders. Government interventions are modeled as part of an overall market for credit, which may be characterized by market clearing, rationing, or redlining of different borrower groups. Thus, it is possible to see how the effects of credit interventions depend upon the initial regime. In addition, the model accounts for the financing costs of federal credit, interactions among programs, and the use of alternative credit instruments.

The main contribution of this paper is to simulate the model described above to generate numerical estimates of the effects of

*Department of Economics, UCLA, Los Angeles, CA 90024-1477. This paper is based on my Ph.D. dissertation. I thank my advisor, Michael Boskin, for many helpful discussions. I have also benefitted from comments from Doug Bernheim, David Butz, Al Harberger, Frank Howland, Seonghwan Oh, Eric Rasmusen, John Riley, Russell Roberts, John Karl Scholz, John Shoven, Jonathan Skinner, Michael Waldman, and two anonymous referees, and I gratefully acknowledge financial support from the John M. Olin Foundation. I thank Lorraine Grams for expert assistance in the preparation of this paper.

[1] All statistics in this paragraph are based on the data provided in Table 1, except for new federal and federally assisted lending, which is based on data provided in "Special Analysis F" of the U.S. Budget.

[2] Bruce Smith (1983) discusses optimal government lending with imperfect information but focuses on monetary policy rather than targeted credit subsidies. An important earlier paper on credit subsidies is Rudolph Penner and William Silber (1973).

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credit programs. Principal conclusions are as follows. Existing credit subsidies raise aggregate private investment by between 0 percent and 4 percent, depending on the elasticity of the supply of funds. Crowding out depends critically on the supply elasticity but is bounded at a maximum of about 5 percent of original borrowing by nontargeted sectors. The allocational effects of lending policies depend on the size of the effective subsidy, rather than on credit volume. As a consequence, lending programs exert important effects on the allocation of funds to farmers, students, small businesses, and tax-exempt borrowers. The well-known federal mortgage-guarantee programs are estimated to have smaller net effects because of the smaller effective subsidies they provide. The allocational effects can also depend on the initial state of the market. If a target group is rationed, credit programs can exert very strong effects, by releasing the rationing constraint.

Although the underlying model allows for the possibility of welfare-improving interventions, the estimated efficiency cost of actual credit policy is high: approximately $10–15 billion, or 1/3 percent of gross national product in 1987. These estimates are relatively insensitive to assumptions about the elasticity of supply of funds. Therefore, even if the crowding-out effects are small, the welfare loss is still sizable.

Because both new and inframarginal borrowers receive funds, credit subsidies cost the government in excess of 50 cents per dollar of *incremental* target-group investment. In addition, most direct welfare gains appear to accrue to borrowers who would have received credit without government subsidies. These subsidies represent pure windfall gains for the recipients, with no obvious corresponding societal benefits. All current programs require the existence of large external benefits to be welfare-improving. Even programs for groups that would have been excluded from private markets without federal assistance (say, students) face this requirement, because the default and subsidy costs are so large.

Because a subsidy to one target group crowds out other target groups as well as nontargeted groups, interactions among programs can indirectly eliminate much or all of the original direct gain. In the low-supply-elasticity case examined here, up to 15 percent of the direct gains to farmers, students, and small businesses are offset by the effects of other credit programs. Interactions can be even more severe if a target group is rationed or redlined.

Two general principles emerge for policy reform. First, the effectiveness of any proposed reform depends on the structure of the original subsidy. For example, the vast majority of the cost of student-loan guarantees stems from the government's willingness to cover interest payments until the student graduates. Many policy proposals, however, focus on reducing the default rate, which would have only a minor impact on the costs of the program. Second, programs in which the government provides appropriate marginal incentives appear to be more efficient than programs that attempt to replace private credit arrangements. These results indicate that, although there is still a role for government in the more marginal sectors of the credit market, that role is more limited and somewhat different in nature than current policies would suggest.

Section I provides an overview of current federal credit activity. Section II develops the formal model. Section III specifies parameter values. Section IV presents the main results. A concluding section discusses caveats and extensions to this line of research. Appendixes A and B describe some important calculations and provide detailed references for the parameter specifications, respectively.

I. An Overview of Federal Credit Subsidies

The government supplies credit through several instruments.[3] Direct loans typically offer large subsidies and are concentrated in the agricultural and rural sectors. Loan

[3]Detailed information concerning federal credit programs may be found in "Special Analysis F," in *Special Analyses: Budget of the U.S. Government* (any year) or in Bosworth et al. (1987).

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TABLE 1—FEDERAL LENDING AND DOMESTIC CREDIT MARKETS (DOLLAR AMOUNTS IN BILLIONS)

Statistic	Fiscal Year								Total
	1980	1981	1982	1983	1984	1985	1986	1987	
Net credit advanced in nonfinancial credit markets	341.7	375.9	388.9	550.2	753.9	854.8	831.7	685.2	4,782.3
Net federal and federally assisted lending	110.2	109.9	131.8	140.2	129.9	246.7	159.9	179.5	1,208.1
Direct loans	24.2	26.1	23.4	15.3	6.3	28.0	11.2	−19.0	115.5
Guaranteed loans	31.6	28.0	20.9	34.1	20.1	21.6	34.6	60.4	251.3
GSE	24.1	32.4	43.3	37.1	53.1	60.7	83.3	107.8	441.8
Tax-exempt credit	30.3	23.4	44.2	53.7	50.4	136.4	30.8	30.3	399.5
Federal borrowing from the public	70.5	79.3	135.0	212.3	170.8	197.3	236.3	151.7	1,253.2
Federal lending as a percentage net credit	32	29	34	25	17	29	19	26	25
Federal lending as a percentage of federal borrowing	156	138	98	66	76	125	68	118	96
Federal lending as a percentage of nonfederal borrowing	41	37	52	41	22	38	27	34	34

Note: Net lending is new lending minus repayments.
Sources: Office of Management and Budget, "Special Analysis F," *Special Analyses: Budget of the United States*, 1988 table F-22; Board of Governors of the Federal Reserve System, *Flow of Funds Accounts*, Fourth Quarter, 1987, pp. 2–3.

guarantees assist a wide range of borrowers, particularly home-owners, small businesses, students, and exporters. Clifford Hardin and Arthur Denzau (1981) report the existence of more than 350 direct and guaranteed lending programs. Government-sponsored enterprises (GSE) aid borrowers in housing, agricultural, and student loan markets, primarily through the operation of secondary markets.[4] Tax-exempt status allows state and local governments to borrow at reduced cost and to operate their own credit programs by passing on the interest savings to preferred borrowers. Effective subsidy rates, default rates, and credit volume vary greatly across sectors and programs and are discussed in detail below.

Credit policies are typically justified in terms of preexisting distortions (e.g., student loans), distributional concerns (farm loans), or externalities (small businesses). Credit subsidies have also been proposed as solutions to issues such as energy development and the maintenance of international competitiveness. In addition, emergency loan guarantees have assisted Chrysler, Lockheed, and New York City in recent years.[5]

As shown in Table 1, lending programs account for an important share of credit advanced in domestic capital markets. The volume of net federal lending has grown at approximately the same rate as overall credit in the 1980's and has exceeded $100 billion in every year since 1980. In a typical year, federal lending accounts for one-fourth of all net credit advanced and one-third of all nonfederal borrowing. Since 1980, the amount of net federal and federally assisted lending is approximately the same as the amount of federal borrowing.

Extensive lending activity, per se, does not imply that credit policy imparts important effects on the economy. However, the sheer magnitude and pervasiveness of credit interventions suggest the need for further

[4]For a detailed discussion of issues concerning government-sponsored enterprises see Thomas Stanton (1988).

[5]Paul Chaney and Anjan Thakor (1985) analyze emergency guarantees. However, these should be distinguished from ongoing guarantee programs that guarantee many loans and smaller amounts per loan and that do so at the time the loan is made.

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analysis. In particular, the volume of federal credit activity indicates that any formal analysis should consider the effects of subsidies on nontargeted groups and the financing of credit policy. The diversity indicates the need to model several types of lending instruments and interactions among credit programs. Finally, since many of the targeted groups appear to be the marginal borrowers in modern credit markets, analysis of both rationing and market-clearing outcomes will be relevant. The next section outlines a model with these characteristics.

II. A Model of the Credit Market[6]

The underlying model is a variant of Stiglitz and Weiss (1981) and analyzes a competitive loan market in which lenders have imperfect information about borrowers. The market consists of government and many borrowers, depositors, and financial intermediaries. All agents are risk-neutral; there is no aggregate risk. Depositors supply funds to the market according to a function $S(\rho)$, where ρ is the certain rate of return on bank deposits.

Borrowers are divided into $n+1$ groups: n target groups for credit policy and one general (nontargeted) group.[7] Borrowers are characterized by two pieces of information: their group identity and their location within that group. Group identity is assumed to be public information and will provide a (noisy) signal of borrowers' riskiness. Location within a group refers to the riskiness of the individual's projects and is known only to the individual borrower. With these informational assumptions, lenders (including the government) can determine which groups are eligible for credit subsidies, yet still face residual uncertainty within each group.

Borrower groups are indexed by $i = 0, 1, \ldots, n$ and characterized by two behavioral assumptions. First, the demand for loans (L_i^D) is a decreasing function of the effective interest rate (r_i^*) paid by the borrower. Second, the repayment rate (ϕ_i) falls as r_i^* rises. That is, increases in effective interest rates generate adverse selection in each loan market.[8]

Financial intermediation ("banking") is subject to free entry and constant returns to scale. Since they can discern only a noisy signal of borrowers' riskiness, banks set different loan rates for each group.[9] The expected return to the bank on a loan to target group i at rate r_i is

$$\rho_i = \rho_i(r_i, \phi_i, \mathbf{C}_i)$$

where $\mathbf{C}_i$ (a vector) represents credit policy for target group i. Holding credit policy constant, the existence of adverse selection implies that an increase in lending rates may eventually lead to a reduction in the banks' expected return.[10] This relationship is shown as the ρ_j curve in Figure 1.

[6]This section provides a brief overview of the underlying theoretical model, which is derived rigorously in Gale (1990b).

[7]Additional nontargeted groups can be incorporated as well.

[8]These results can easily be derived from primitive assumptions on information, endowments, technology, and preferences. For example, Gale (1990b) uses the information structure above and assumes that an investor in group I with riskiness j has no endowment, must borrow to invest, has an available project of unit size which succeeds with probability $p_I(j)$, pays $R_I(j)$ if successful, and pays 0 otherwise. For any two borrowers j_1 and j_2 in group I, if $R_I(j_1) > R_I(j_2)$ then $p_I(j_1) < p_I(j_2)$. This implies that

$$L_I^D(r_I^*) = \int_0^{j_I^*(r_I^*)} f_I(j)\, dj$$

and

$$\phi_I(r_I^*) = \left[\int_0^{j_I^*(r_I^*)} p_I(j) f_I(j)\, dj\right] \Big/ F_I(j_I^*(r_I^*))$$

where j_i^* is the marginal borrower in group j (i.e., j_i^* is defined by $R_i(j_i^*) = r_i^*$) and where f and F represent the density and distribution of j, respectively. Similar specifications can be used to generate the supply curve. Thus, the underlying supply and demand functions are structural (i.e., invariant with respect to credit policy).

[9]Allowing banks to set an interest and collateral requirement for each group would not change the results as long as the collateral is incomplete (i.e., as long as banks prefer that borrowers repay rather than default).

[10]More generally, adverse selection implies that ρ need not be monotonic in r (see Stiglitz and Weiss, 1981).

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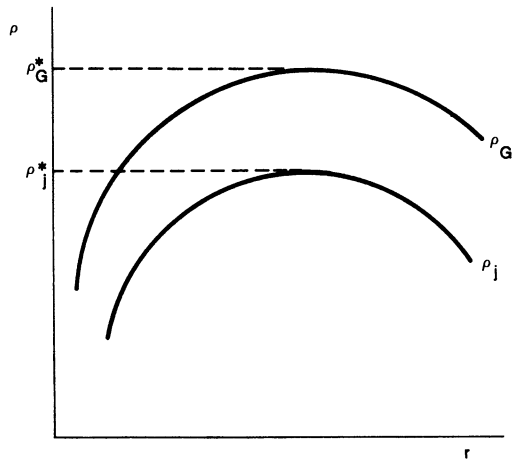


FIGURE 1. BANK RETURN ON GENERAL LOANS (ρ_G) AND TARGET-GROUP LOANS (ρ_j)

As shown in Figure 1, the maximum bank return on general loans (ρ_G^*) is assumed to be at least as great as that available on target group loans: $\rho_G^* \geq \rho_j^*$, $j = 1, \ldots, n$.[11] This assumption will prove necessary and sufficient to imply that situations exist in which the general market clears while target groups are rationed or redlined.

The government may assist target groups through subsidized lending, loan guarantees, or tax-exempt status. The subsidies reduce the effective interest rate and raise the expected bank return to target-group lending indirectly through $\phi_i(r_i^*)$ and in some cases directly (e.g., loan guarantees). The government borrows to fund its credit programs and pays its creditors with program revenue first and lump-sum taxes on depositors. These assumptions force explicit recognition of the costs of the programs. The government has the same information and borrowing costs as banks.

The model is competitive in that there are many agents and entry is free. There is, however, one important departure from the standard competitive framework: in addition to caring about aggregate demand, banks also care about the identity of the borrowers demanding loans. Thus, an important concept will be effective demand —demand for loans that banks are willing to supply. Effective demand for borrower group i is given by

$$L_i = \begin{cases} L_i^D & \text{if } \hat{\rho} < \rho_i^*(r_i^*) \\ 0 & \text{if } \hat{\rho} > \rho_i^*(r_i^*) \end{cases}$$

where $\hat{\rho}$ is the equilibrium value of ρ. For example, in Figure 1, if the equilibrium cost of funds is greater than ρ_j^*, banks will not consider making loans to group j, since such loans would generate negative expected profits. If $\hat{\rho} = \rho_i^*$, L_i is determined as a residual, after other credit demands have been met, such that $0 \leq L_i \leq L_i^D(r_i^*)$. These considerations imply that aggregate effective demand, shown in Figure 2 (for $n = 1$), is a step function.[12] In equilibrium, the market for target-group loans may be characterized by market clearing (S_3), rationing (S_2), or redlining (S_1), depending on the relative magnitudes of supply and effective demand.[13]

Equilibrium is characterized by two economic conditions. First, banks' expected returns to lending are equalized across all groups that receive loans and are equal to the cost of funds (zero profits):

$$(1) \qquad \hat{\rho} = \rho_i \qquad \text{if } \rho_i^* \geq \hat{\rho}.$$

Second, the sum of private and public demands equals the supply of funds:

$$(2) \qquad S(\rho) = \sum_{i=0}^{n} [L_i + G_i]$$

where G_i is government borrowing to fund programs for group i. A third set of equations shows that credit subsidies drive a

[11] Figure 1 also shows $\rho_G(r) > \rho_j(r)$ $\forall j, r$. This assumption is not necessary for any of the results, but simplifies the figure.

[12] See John Riley (1987) for an independent derivation of this demand curve.

[13] A supply curve intersecting the horizontal demand strip to the left of S_1 would represent rationing of general borrowers and redlining of target borrowers. This case is not considered.

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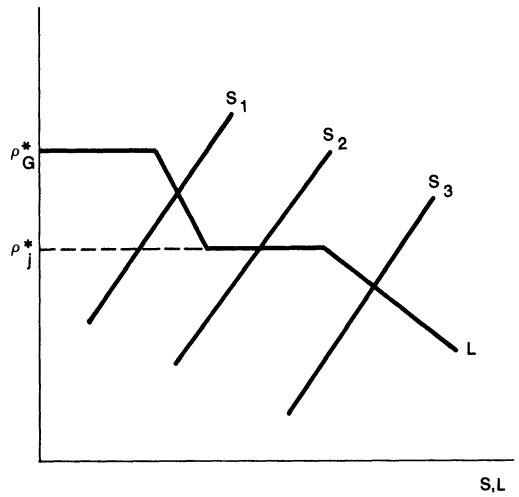


FIGURE 2. AGGREGATE EFFECTIVE DEMAND FOR $n = 1$

wedge between effective interest rates and loan rates:

$$(3) \quad r_i^* = g(r_i, \mathbf{C}_i) \qquad i = 0, 1, \ldots, n.$$

The simulations employ the following supply and demand functions:

$$(4) \quad S = a(\rho)^{\alpha}$$

$$(5) \quad L_i^{\mathrm{D}} = b_i(r_i^*)^{\beta_i} \qquad i = 0, 1, \ldots, n$$

where α and β_i represent constant elasticities and a and b_i are constants. As shown in Appendix A, government borrowing to fund credit policy for group i is given by

$$(6) \quad G_i = L_i s_i$$

where s_i is the percentage reduction in the present discounted value of borrowers' loan payments due to federal credit subsidies.

Appendix A also shows that the relationship among ρ, r_i, and r_i^* may be reduced to a single set of equations, summarized here as[14]

$$(7) \quad r_i^* = f(\rho, \phi_i, C_i) \qquad i = 0, 1, \ldots, n.$$

[14]The loan rate, r_i, vanishes from the equilibrium calculation because it is determined by ρ, ϕ_i, and C_i.

The effective interest rate rises with increases in deposit rates and falls with increases in credit subsidies and the probability of repayment. Equations (2) and (4)–(7) represent the market-clearing equilibrium, with S, L_i^{D}, G, ρ, and r_i^* endogenous.[15]

If group j is rationed, equation (2) is modified to become

$$(8) \quad L_j + G_j = S - \sum_h \left(L_h^{\mathrm{D}} + G_h\right)$$

where h indexes groups with $\rho_h^* > \hat{\rho}$. Lending to group j is determined as a residual. As shown in Figure 2, the equilibrium cost of funds must equal the maximum bank rate of return on group-j loans:

$$(9) \quad \hat{\rho} = \rho_j^*.$$

Equations (4)–(9) represent the rationing equilibrium.

One important characteristic of the model is that, in equilibrium, banks order borrower groups by their maximum rate of return and serve the groups sequentially. If the equilibrium cost of funds is $\hat{\rho}$, all groups j with $\rho_j^* > \hat{\rho}$ have clearing credit markets, those with $\rho_j^* = \hat{\rho}$ are rationed,[16] and those with $\rho_j^* < \hat{\rho}$ are redlined. Because banks order the borrower groups and because $\rho_{\mathrm{G}}^* > \rho_j^*$, target groups are the residual borrowers. As a consequence, if a target group is rationed, the marginal effects of changes in supply or demand fall completely on the rationed group, while general borrowers are left unaffected. An interesting implication of this result occurs when one target group is rationed and another is redlined: the burden of subsidies to the previously redlined group falls entirely on the rationed group. In this case, credit subsidies simply rearrange loans among target groups; aggregate targeted borrowing does not rise.

A second important feature of the model is that the ordering of projects by borrower returns and social returns will differ, in gen-

[15]Redlining equilibria are identical to market-clearing equilibria, except that the index runs over only those groups with positive loan values.

[16]There is an infinitesimal chance that these groups clear or are redlined, depending on the position of S_2 in Figure 2.

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TABLE 2—CHARACTERIZATION OF FEDERAL CREDIT POLICIES, 1980–1987

Sector	Principal credit instrument	s	s^G	s^O	γ	δ	σ
Mortgage	loan guarantees	0.02	< 0.01	0.02	1.0	1.0	0.01
Farm	direct lending	0.25	0.00	0.25	0.0	0.0	0.00
Student	loan guarantees	0.32	0.04	0.28	1.0	1.0	0.39
Small business	loan guarantees	0.14	0.09	0.00	0.9	0.0	0.91
Tax-exempt	interest subsidy	0.19	0.00	0.19	0.0	0.0	0.00

Variable	Definition
s	overall reduction in the present discounted value of loan payments induced by federal credit subsidies $= s^O + s^G$
s^G	reduction in the present discounted value of loan payments induced by just the subsidized guarantee
s^O	reduction in the present discounted value of loan payments due to other factors
γ	proportion of principal and interest covered by the guarantee
δ	proportion of guarantee fee paid by the borrower
σ	subsidized portion of the guarantee fee

Sources: See text and Appendix B.

eral. As a consequence, subsidies can raise or reduce welfare (defined as total surplus), depending on the investment response and the size of the subsidy (Gale, 1990b). Whether actual policies raise welfare is examined below.

III. Parameter Specification

The specification of federal credit programs is potentially complicated due to the large number and diversity of credit programs, as discussed in Section I. However, a sizeable percentage of both the credit volume and the subsidies inherent in credit programs is concentrated in a relatively small number of programs and sectors. In 1986, the five sets of programs listed in Table 2 accounted for 93 percent of all new credit advanced, subsidized, or guaranteed by the government, including 75 percent of direct loan obligations and 90 percent of guaranteed loans.[17,18] The subsidies in these programs accounted for approximately 75 percent of all subsidies extended through loan guarantees and 50 percent of subsidies extended through direct lending (excluding foreign military sales credit). In order to capture the main effects of credit interventions without needlessly complicating the analysis, the simulations focus on these programs. Table 2 specifies their underlying parameters. A brief discussion is presented here; detailed references are given in Appendix B.

The overall percentage reduction in the present discounted value of a borrower's loan payments induced by a federal subsidy is given by s and is estimated by the Office of Management and Budget.[19] All target groups except mortgage borrowers receive substantial subsidies from the government. These benefits may appear in one of two forms: a subsidized fee for a loan guarantee (s^G) or "other" forms (s^O), including reduced interest payments, deferral of payments, grace periods, and longer maturities. A loan-guarantee fee is considered to be subsidized if it does not cover the expected

[17]These figures omit government-sponsored enterprises (GSE's) other than the Farm Credit System, which are also omitted from the simulations, in order to avoid the need to model the operation of secondary markets. However, since GSE's are concentrated in mortgage, farm, and student credit markets, the importance of credit subsidies to the five sectors listed would increase if GSE's were included.

[18]The programs listed in Table 2 are in some cases meant to refer to combinations of individual programs, as discussed in Appendix B.

[19]Except for tax-exempt borrowers, estimates may be found annually in "Special Analysis F," *Special Analyses: Budget of the United States*. Although the estimates are necessarily rough, they serve to provide a general impression of the overall subsidy rate. The calculation of the overall subsidy for tax-exempt borrowers is provided in Appendix B.

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TABLE 3—SPECIFICATION OF OTHER PARAMETERS

Sector	L	β	ϕ	r^*	$\hat{r}$
Mortgage	4.6	−1.80	0.97	0.101	0.103
Farm	2.0	−1.00	0.88	0.078	0.118
Student	0.6	−0.65	0.91	0.032	0.122
Small Business	0.2	−0.80	0.90	0.092	0.127
Tax-Exempt	8.3	−0.40	1.00	0.054	0.099
General Borrowers	81.9	−0.80	1.00	0.100	0.099

Variable	Definition
L	credit allocation as percentage of total credit
β	interest elasticity of loan demand
ϕ	repayment rate
r^*	effective interest rate
$\hat{r}$	interest rate in the absence of credit subsidies

Notes: Private credit allocations (L) sum to only 97.6, rather than 100, because government for credit subsidies requires 2.4 percent of available credit. The parameters r^* and $\hat{r}$ were calculated on the assumption that $\rho = 0.10$.
Source: See text and Appendix B.

default costs. The composition of the overall subsidy varies considerably across programs.

The parameters used to calculate s^G and s^O are listed next: γ is the guarantee rate, and δ is the proportion of the fee paid by the borrower; both are obtained from government documents. The percentage of default costs not covered by the guarantee fee is given by σ and is calculated in Appendix B.

Table 3 presents estimates of other key parameters. Credit allocations are normalized to equal 100 and are based on recent historical averages. The allocations for each target group are meant to include only federal lending. Consequently, unsubsidized loans (including those for mortgages, farms, etc.) are included in general borrowing. Demand elasticities are obtained from recent studies.

Repayment rates are specified as constants, rather than functions of interest rates (as in Section II), for simplicity and due to the lack of any useful data on the shape of the function. It is still possible to generate rationing outcomes by specifying a maximum effective interest rate for each group and assuming that banks will not lend to the target group at higher rates. The effect of this assumption is to generate a $\rho - r$ curve that rises linearly to an interior maximum at r_j^* and then falls. Thus, the specification of constant ϕ's considerably simplifies the calculations without losing any important aspect of the underlying model. For general borrowers, ϕ is set to 1.0. Other reported repayment rates are based on government data and existing studies.

Effective interest rates (r^*) are calculated (as described in Appendix A) on the assumption of $\rho = 0.10$. Estimated interest rates in the absence of credit subsidies are presented in the last column. Credit subsidies lower effective target-group interest rates below those paid by general borrowers for all target groups except mortgage borrowers.

Perhaps the single most important and controversial parameter is the elasticity of supply of funds, or more generally the openness of capital markets. A large literature on this topic has developed from papers by Arnold Harberger (1978, 1980) and Martin Feldstein and Charles Horioka (1980), with mixed conclusions. The supply of credit from domestic sources is also controversial. Robert DeFina (1984) surveys existing results and lists interest elasticities ranging from 0 to 5. This paper employs two alternative assumptions concerning the elasticity of supply: $\alpha = 5.0$ and $\alpha = 0.5$. Using

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these extremes will provide bounds for the effects of credit policies and will help indicate which results are robust to alternative views about capital markets.

IV. Results

Before discussing the effects of credit policies, it is necessary to resolve two further issues: 1) the interpretation of existing credit allocations (summarized in Table 3) as representing market-clearing or rationed outcomes and 2) assumptions concerning the allocation of credit without any lending subsidies. I specify two interpretations of existing credit patterns and policies: one in which all groups clear and one in which farmers are rationed but other groups clear.[20] The allocation of credit without lending subsidies depends on assumptions about whether the market is rationed with existing credit policies, as well as assumptions concerning the shape of the $\rho - r$ curve for each group. If all markets clear with existing policies, I examine two scenarios in the world without credit subsidies: (a) all markets clear and (b) tax-exempt and mortgage markets clear, but farmers, students, and small businesses are redlined. These cases are meant to represent optimistic and pessimistic extremes concerning the viability of target-group markets in the absence of credit policy. If the existing credit allocation represents rationing of farmers, I assume that case (b) holds without credit policy.

A. *Allocation and Level of Credit*

The effects of credit subsidies on the allocation and level of credit are presented in Table 4. Starting from a world with no credit subsidies, as represented by scenario (a) or (b) described above, the introduction of all credit programs generates the base-case allocations listed in Table 3 (reproduced here for convenience). Many of the aggregate effects depend on the elasticity of supply.

In the high-elasticity case ($\alpha = 5$), credit programs raise private borrowing by between 2.5 percent [scenario (b)] and 4.1 percent [scenario (a)]. Crowding-out effects are very small: general borrowing falls by approximately 1 percent in either scenario. Aggregate target-group investment increases by between 23 percent and 42 percent. With the exception of mortgage borrowers, all target groups experience substantial increases in investment.

If the supply of funds is inelastic ($\alpha = 0.5$), different aggregate patterns emerge. Credit programs have essentially no net effect on aggregate private borrowing. General borrowing falls by between 3.5 percent in scenario (a) and 4.6 percent in scenario (b) but is matched by an equivalent absolute rise in targeted lending.

Table 4 indicates that credit programs exert powerful effects on the allocation of credit to target groups, regardless of the supply elasticity. Other simulations (not reported) indicate that these sectoral effects remain for a wide range of demand elasticities as well.

B. *Welfare Effects*

Standard welfare analysis (see Harberger, 1971) aggregates consumers' and producers' surplus and assumes that the government has access to lump-sum taxes. The welfare analysis described here is based on the traditional approach, with two qualifications. First, consumers' surplus is multiplied by the probability of repayment, so that borrowers obtain the surplus only if their project succeeds. Second, loan demand is assumed to become flat at $r^* = 50$ percent and to be zero above that point. This affects the level of consumers' surplus but will not affect the difference in welfare effects across policies, unless a group is redlined.

The simulations indicate that credit programs generate sizable efficiency costs. Using 1980–1987 averages for net credit and gross national product (GNP), credit programs are estimated to reduce welfare by 0.27–0.40 percent of GNP. Using 1987 val-

[20] I assume that farmers are rationed at 80 percent of their desired quantity of loans. It is straightforward to analyze the effects of rationing on other groups as well. The important analytical point, however, is how effects differ across rationing and clearing regimes, which may be illustrated with any single group.

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TABLE 4—ALLOCATION OF FUNDS UNDER ALTERNATIVE SCENARIOS

Sector	Base case	$\alpha = 5.0$ (a)	$\alpha = 5.0$ (b)	$\alpha = 0.5$ (a)	$\alpha = 0.5$ (b)
Mortgage	4.60	4.48	4.50	4.75	4.87
Farm	2.00	1.31	0	1.36	0
Student	0.60	0.25	0	0.26	0
Small business	0.20	0.15	0	0.16	0
Tax-exempt	8.30	6.50	6.51	6.58	6.62
Total, target groups	15.70	12.69	11.01	13.10	11.49
General borrowers	81.91	82.55	82.76	84.78	85.72
Total, all groups	97.61	95.24	93.77	97.88	97.21
ρ	0.1000	0.0990	0.0987	0.0958	0.0945

Notes: Scenarios are described in the text. For the base case, government borrowing to fund credit programs = 2.39, so total credit sums to 100; for the alternative cases (a) and (b), government borrowing is zero.

TABLE 5—GOVERNMENT COSTS ($) PER DOLLAR OF INCREMENTAL TARGETED INVESTMENT

	Demand elasticities					
	High		Original		Low	
Sector	(a)	(b)	(a)	(b)	(a)	(b)
Mortgage	0.54	0.66	0.74	0.92	1.29	1.70
Farm	0.53	0.25	0.72	0.25	1.31	0.25
Student	0.43	0.32	0.55	0.32	0.91	0.32
Small business	0.44	0.14	0.62	0.14	1.16	0.14
Tax-exempt	0.62	0.62	0.87	0.88	1.64	1.65
Overall:	0.57	0.43	0.79	0.51	1.44	0.63

Notes: All of the estimates assume that $\alpha = 5.0$. Original demand elasticities are given in Table 3; high (low) elasticities raise (reduce) the original figures by 50 percent.

ues, the welfare loss is 0.25–0.37 percent of GNP, or approximately $10–15 billion.[21]

The estimates vary somewhat due to assumptions concerning supply and demand elasticities and initial scenarios. Nevertheless, several aspects of the results warrant comment. First is the sheer magnitude: at 1/3 percent of GNP, the welfare loss due to credit programs is substantial. Second, since the results are fairly insensitive to specification of supply or demand elasticities, it is not possible to infer the efficiency costs from the allocational effects. In particular, the absence of crowding out implies only that the costs are borne by agents other than nontargeted investors.

Third, the results indicate the importance of explicitly considering the financing costs of credit programs. For most simulations, the gains and losses to the various borrower groups and suppliers of funds roughly offset each other. *The welfare loss occurs because the programs must be financed.* As shown in Table 5, government costs per dollar of incremental targeted investment are ex-

[21]These numbers are generated by multiplying the welfare changes when the total credit allocation is 100 by (actual net credit)/100 and dividing by GNP. If demand curves do not become flat until $r^* = 100$ percent, the efficiency cost falls by approximately 5 percent. If resource costs of distortionary taxation are included at 30 cents per dollar of government revenue (see Charles Ballard et al., 1985; Kenneth Judd, 1987), the efficiency cost rises to 0.37–0.55 percent of GNP.

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traordinarily high.[22] Using $\alpha = 5$ and the original demand elasticities, the costs exceed 50 cents for each group when starting from scenario (a). For scenario (b), the costs are somewhat lower because several groups are redlined initially, but the average cost is still 50 cents. These results are not sensitive to shifts in α. However, increasing demand elasticities by 50 percent raises the average cost to over 90 cents for any group that is not redlined beforehand.

In a perfectly efficient world, the traditional approach would provide all of the necessary information for welfare analysis. However, focusing solely on the aggregate welfare effects may be misleading because, as discussed above, credit programs are often enacted specifically in response to some preexisting distortion, externality, or distributional concern. Although it is beyond the scope of this paper to develop a model explicitly incorporating all of these issues, several useful insights concerning the welfare effects of credit policy can be derived.

First, under all scenarios, all target groups except mortgage borrowers experience large gains in consumer surplus. However, because the programs subsidize both inframarginal and incremental borrowers and because there are no societal benefits to subsidizing the former group, the division of direct welfare benefits should be an important policy concern.

Under scenario (a), the vast majority of direct welfare benefits accrue to target-group members who would have received credit without public assistance. Approximately 76 percent of student welfare gains are inframarginal. For farmers, small businesses, and tax-exempts, 90 percent of the benefits constitute lump-sum transfers. More than 99 percent of mortgage borrowers' welfare gains also fall into this category. Under scenario (b), of course, none of the gains made by farmers, students, or small businesses are inframarginal.

Therefore, whether credit programs principally aid inframarginal borrowers or new borrowers depends on assumptions concerning the status of sectoral credit markets without lending subsidies. Because credit policy has been an integral part of domestic credit markets for many years, it is difficult to determine with precision which scenario, (a) or (b), is closer to reality.[23] However, whenever the target market clears, any *marginal* increase in subsidies will benefit primarily inframarginal borrowers.

A second issue is the existence of external benefits. Although specification and calculation of external benefits would be extremely complicated, it is possible to calculate how large the benefits would need to be to offset the welfare losses described above. For each program, it is convenient to express the minimum level of external benefits needed to raise welfare as a percentage of the change in the target group's investment. These percentages are called threshold levels and are shown in Table 6.[24]

Sizable external benefits must occur for current policies to raise welfare. Using the original demand elasticities and $\alpha = 5$, external benefits must be 40–70 percent of the change in target-group investment, if the target-group market clears beforehand. Thresholds for marginal changes in policy are slightly lower than for overall changes. Threshold levels for redlined groups are considerably lower, because the change in target-group investment is much larger. For

[22] These figures are calculations of the model, given the definition of government borrowing in equation (6). The numbers bear no relation to any reported budget figure. Federal budgetary treatment of credit programs is highly misleading and inconsistent (see Bosworth et al. [1987], the Congressional Budget Office [1984] or Gale [1990a] for discussion).

[23] It is not obvious that students (who have no collateral or credit history) and small businesses (who must be turned down by at least two banks to qualify for a guaranteed loan) would be redlined in the absence of public assistance. For example, parents could borrow on behalf of their children. SBA guarantees are typically for larger loans with longer maturities than private credit to small business. Without guarantees, borrowers could perhaps qualify for shorter-term or smaller loans. In addition, under current SBA regulations, lenders have incentives to reject private applications in hope of acquiring a guarantee.

[24] Since the estimates assume that investment crowded out by a credit policy contains no external benefits, the threshold levels are lower bounds on the level of external benefits required to raise welfare.

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TABLE 6—THRESHOLD LEVELS (PERCENTAGE) FOR ALTERNATIVE CREDIT POLICIES

Sector	Target-group market with no credit subsidy	Proposed policy	Demand elasticity		
			High	Original	Low
Mortgage	clearing	current policy	30	45	90
	clearing	increase in subsidy	31	46	90
Farm	clearing	current policy	52	70	125
	redlined	current policy	13	14	41
	clearing	increase in subsidy	55	68	109
Student	clearing	current policy	41	51	129
	redlined	current policy	26	21	11
	clearing	increase in subsidy	43	47	60
Small business	clearing	current policy	38	57	93
	redlined	current policy	3	~ 0[a]	−9
	clearing	increase in subsidy	40	52	89
Tax-exempts	clearing	current policy	55	77	143
	clearing	increase in subsidy	50	66	108

Notes: Threshold levels are defined in the text. Estimates use $\alpha = 5.0$. Original demand elasticities are provided in Table 3; high (low) elasticities raise (reduce) the original estimates by 50 percent.

[a]Between 0 and −0.1.

farmers and students, threshold levels fall to 14 percent and 21 percent, respectively. Although these figures may appear to be encouragingly low, they are noteworthy primarily because they are still positive: even if the government creates the market for farm or student loans (with current policies), there is a welfare loss unless sufficient *external* benefits exist. In contrast, SBA guarantees may generate welfare gains even in the absence of external benefits.[25]

Threshold levels are not sensitive to shifts in α but rise dramatically for lower demand elasticities. Even in the most favorable case of high demand elasticities, threshold levels are on the order of 30–50 percent. Credit programs may generate external benefits of the appropriate magnitude to raise welfare. However, in the absence of any compelling empirical evidence that such benefits occur, the results serve primarily to emphasize the efficiency losses associated with credit interventions.

C. *Interactions and the Effects of Credit Rationing*

Because credit subsidies reallocate funds and raise the overall rate of return, a program that subsidizes one target group will necessarily affect credit allocations to other groups. The magnitude of the consequent crowding out is clearly an important policy concern. If policies completely or largely offset each other, then the government is spending substantial resources for little net gain.[26]

Assume that, in the absence of credit programs, target group j receives $D_j(0)$ in funds. When only a subsidy for group j is introduced, the group receives $D_j(j)$. When all current programs exist, group j receives $D_j(\text{all})$. The offset is then measured by:

$$\text{OFFSET}_j = \frac{D_j(j) - D_j(\text{all})}{D_j(j) - D_j(0)}.$$

That is, the offset is the percentage of the gain induced by the original credit subsidy that is eliminated by the introduction of all other credit programs. Estimates of offset percentages are shown in Table 7.

Starting from scenario (a) and setting $\alpha = 5$ yields very small interactions for all credit programs except mortgages. Almost two-fifths of the original rise in mortgage credit due to subsidies is eliminated by other

[25]These results do not depend on the assumption that demand becomes flat at $r = 50$ percent.

[26]The importance of such interactions has been recognized independently by Bosworth et al. (1987). They do not quantify these effects, however. Penner and Silbert (1973) focus on interactions among programs that focus on the same target group.

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TABLE 7—OFFSETS: THE EFFECTS OF INTERACTIONS AMONG PROGRAMS

Sector	Scenarios with no subsidy	Interpretation of base case	Offsets (percentage) $\alpha = 5.0$	Offsets (percentage) $\alpha = 0.5$
Mortgage	(a)	clearing	38.3	175.0
Farm	(a)	clearing	2.5	10.9
	(b)	rationed[a]	33.8	35.3
	(b)	rationed[b]	17.4	19.8
Student	(a)	clearing	2.6	11.0
Small business	(a)	clearing	3.5	14.9
Tax-exempt	(a)	clearing	1.2	4.9

Notes: Offsets are defined in the text. Estimates shown are based on the demand elasticities reported in Table 3.

[a]Rationed at 67 percent of notational demand.

[b]Rationed at 83 percent of notational demand.

programs. This large crowding-out effect occurs because the original subsidy to mortgages is small ($s = 0.02$). Thus, even a small increase in ρ eliminates a substantial percentage of the original benefits. For the other groups, the offset is small because direct subsidies are large.

At the lower value for α, interactions are much larger. For groups other than mortgage borrowers, the interactive effect quadruples: 15 percent of small-business investment gains and 10 percent of increases in farm and student credit are eliminated by other federal credit programs. Although these interactions are unlikely to have important macroeconomic effects, their effects on individual sectors are nonetheless sizable. For mortgage borrowers, the results are even more striking. They would be better off if all subsidies were eliminated. The crowding out induced by other subsidies more than offsets the small direct subsidy provided by government. Thus, as shown in Table 4, mortgage credit falls with the creation of all programs.

Interactions become extremely important when a group (e.g., farmers in Table 7) is rationed given current credit policies. If only a farm policy is introduced from scenario (b), farm credit rises to a point on farmers' notional demand curve. However, as other credit programs are introduced, interest rates rise and make the farmers' rationing constraint bind. Farmers' notional demand at that rate is typically much higher than the residual amount allotted to them. If farmers are rationed at 67 percent (83 percent) of their notional demand, the offset is approximately 34 percent (18 percent). In any case, if a group is rationed in equilibrium, the offsets will typically be large, regardless of α. Offsets are not sensitive to variations in demand elasticities.

D. *Policy Reform*

This section uses the model to examine the implications of some recent events and proposals to reform federal credit. Because specific features of individual credit markets may be important in this regard, the results are necessarily exploratory.

Due to the high and rising costs of federal farm credit, many proposals have focused on reducing the subsidy rate. Halving the rate would reduce farm borrowing by 22 percent and would reduce government costs by 60 percent. Elimination would reduce farm credit by 35 percent.

Sharp increases in observed defaults at the FHA have directed a considerable amount of attention to modifying mortgage loan guarantees. If the overall default rate on FHA guarantees ultimately rises to 10 percent (from the 3 percent used above to represent earlier years), the welfare costs of mortgage guarantees would rise fourfold.[27] However, mortgage guarantees would still be the *least* subsidized of the five major credit programs analyzed here.

[27]The results presented in this section are based on $\alpha = 5$, but are not sensitive to changes in α.

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Although they occupy only a small portion of modern credit markets, guaranteed loans to students and small businesses have generated a substantial amount of controversy, presumably because of their perceived high costs. However, because of the differing structure of the programs, different methods for reducing costs are warranted. For example, as shown in Table 2, the principal subsidy to students arises from the stipulation that borrowers may defer principal and interest payments until nine months after graduation with no penalty. Elimination of that subsidy reduces the government's costs by 96 percent and reduces loan demand by 60 percent. However, most of the policy debate has focused on collecting defaulted student loans. Eliminating government costs from default (i.e., setting the guarantee fee equal to the loss rate) would reduce government costs by only 23 percent and would reduce demand by 14 percent.

In contrast, most of the benefits of SBA loan guarantees reside in the subsidized guarantee fee. Setting the guarantee fee equal to the loss rate would reduce small business borrowing by 16 percent, but government costs would fall by 70 percent. Thus, defaults are an important issue for SBA loan guarantees, whereas for students, program costs will continue to be large as long as deferral until graduation is allowed.

In its Preferred Lenders' Program, the SBA offers lending institutions significant reductions in paperwork and allows for greater lender discretion in making loans in exchange for reducing the guarantee rate to 75 percent from 90 percent. If the program works as intended, the increased flexibility will attract more lenders to the program, while the reduced guarantee will induce more careful screening. The effect of the program should be to reduce loss rates. If loss rates fall by 2.5 percentage points (from 10 percent to 7.5 percent), the allocation of funds to small business would fall by 1.5 percent, but small business welfare would *rise*, because the probability of repayment increases. Government borrowing for small business would fall by 30 percent and aggregate welfare would *rise* slightly.[28] Although this result is speculative, the analysis indicates the potential for credit programs to raise welfare by improving the incentives and organization of sectoral credit markets, while still preserving the basic purpose and nature of the credit policy.

V. Conclusions

The role of government in imperfect capital markets is an issue of obvious theoretical and practical concern. This paper has developed a model of the credit market and simulated the effects of federal credit interventions. Major results are reported in the introduction. Some caveats and possible extensions should be mentioned at this stage. First, because any simulation employs a variety of maintained assumptions and approximations, results should be interpreted as showing the direction and relative magnitude of various effects, rather than as precise estimates.

Second, since the model assumes that all agents are risk-neutral, all assets are perfect substitutes, and rates of return are equalized across assets. However, many assets may not be good substitutes in asset demand (Benjamin Friedman, 1978; Jeffrey Frankel, 1985). In the present context, credit for students, small businesses, and farmers may be poor substitutes for mortgages, tax-exempts, or corporate debt. In that case, if lenders are risk-averse, the crowding-out effect of subsidies aimed at the first set of target groups would be smaller than indicated above, but so would the direct benefits (Penner and Silber, 1973; Friedman, 1978). With a range of substitutabilities among assets, credit policies could actually crowd in investment by other groups (Friedman, 1978). However, given that 75 percent of all net credit advanced is funneled through financial intermediaries (Michael Moran, 1985), which presumably face competitive pressures to maximize profits, the

[28]Although screening costs would increase, they would at least partially be offset by reduced lender paperwork for the SBA.

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assumption of risk neutrality may be a good approximation.[29]

Sensitivity analysis on a broader scale would consider using an alternative theoretical framework. For example, Stephen Williamson (1987) and de Meza and Webb (1988) rely on monitoring or screening costs to generate imperfect markets. However, simulations of such models may prove difficult, due to the absence of any reliable estimates of those costs. Although de Meza and Webb (1987) show that allowing equity finance can fundamentally change the results derived from the Stiglitz-Weiss framework, borrowers served by federal credit typically do not issue external equity, so eliminating equity contracts does not appear to be a major restriction in this case.

In light of these results, several features of the model presented above make it an excellent candidate for simulations: 1) it allows comparison of market-clearing, rationing, or redlining outcomes; 2) results depend on parameters for which a fair amount of empirical information is available, rather than on screening or monitoring costs; 3) it easily accommodates multiple borrower groups; and 4) welfare effects depend on the particular parameter values, rather than being predetermined by theoretical factors.

Even if alternative models were used, my conjecture is that the basic results would stand: credit policy has important allocational effects, due to the large subsidies provided, and sizable efficiency costs, due to high government-financing costs. These aspects of credit policy appear to drive the results and would have to be incorporated in any theoretical structure.

Perhaps the most important and most difficult extension concerns the connection between credit allocations and real economic activity. The model maintains a simple, direct link: an increase in credit leads to increased real activity. D. C. Rao and Ira Kaminow (1975), Mary Kay Plantes and David Small (1981), and Bosworth et al. (1987) emphasize that subsidized credit may simply induce borrowers to substitute debt for equity or capital for labor. The borrower may also use the funds for some entirely unrelated purpose. Under this view, changes in credit may not induce any shift in real activity.

Alternatively, if it provided the marginal source of funds for a large project, government may raise private investment by more than the government credit extended. In addition, a subsidy may serve to keep an enterprise viable now and raise investment in the future. Under these scenarios, investment rises by more than the change in government credit.

This paper makes no attempt to resolve these issues. Instead, the one-to-one link between credit and investment may be taken as a benchmark. Future research on federal credit may help to clarify real-financial linkages, perhaps through empirical analyses of individual programs. Another issue is how the design of credit programs influences their effects. It is easy to show that guarantees and subsidies generate differing incentive effects (Gale, 1987). Focusing on incentives can explain some of the divergent default behavior across programs, which were simply taken as given above.

Credit programs may also help explain long-term declines in saving. Student and mortgage guarantees reduce the level of initial wealth required to make sizable purchases of education and housing and thereby reduce one motive to save. Fumio Hayashi et al. (1987) find limited support for this proposition in comparing U.S. and Japanese saving data. Finally, it may be possible to exploit changes in credit programs to test for the nature and importance of liquidity constraints.

Appendix A: Model Calculations

Effective Interest Rates

Consider a \$1 loan and assume that all defaults occur in the first period of repayment. Given the probability of repayment, ϕ, the costs of funds, ρ, and the maturity period for loans, m, banks require constant

[29] However, given the presence of federal deposit insurance, banks may be risk-seeking.

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annual loan payments, X_1, such that

$$(A1) \quad X_1 = \left[\phi \sum_{k=1}^{m} (1+\rho)^{-k}\right]^{-1}.$$

The implied interest rate, r_1, is defined by

$$(A2) \quad X_1 = \left[\sum_{k=1}^{m} (1+r)^{-k}\right]^{-1}.$$

Substitution of (A1) into (A2) yields

$$(A3) \quad \sum_{k=1}^{m} \left[(1+\rho)^{-k}\right] = \phi^{-1} \sum_{k=1}^{m} \left[(1+r_1)^{-k}\right].$$

Equation (A3) describes the determination of effective interest rates, given ρ, ϕ, and m, in the absence of credit subsidies. If the government introduces a loan guarantee, the bank now requires annual payments, X_2, such that its discounted costs equal its discounted expected receipts, or

$$(A4) \quad \phi X_2 \sum_{k=1}^{m} \left[(1+\rho)^{-k}\right] + (1-\phi)\gamma = 1 + (1-\delta)(1-\sigma)(1-\phi)\gamma.$$

The left side represents expected bank revenues. The last term on the right represents bank payments for the guarantee fee: the fair-insurance cost of the guarantee is $(1-\phi)\gamma$, but the government pays σ, and borrowers pay δ.

The percentage reduction in the present value of loan payments induced by the subsidized loan guarantee, s_G, is found by solving

$$(A5) \quad 1 - s_G = \frac{\delta(1-\sigma)(1-\phi)\gamma + \sum_{k=1}^{m} X_2/\left[(1+r_1)^{-k}\right]}{\sum_{k=1}^{m} X_1/\left[(1+r_1)^{-k}\right]}.$$

This expression gives the ratio of borrower payments with and without the guarantee. Using (A2) and some algebra, (A5) reduces to

$$(A6) \quad s_G = \sigma(1-\phi)\gamma.$$

Define X_3 as annual borrower payments with guarantees and other subsidies, and define s as the percentage reduction in the present value of loan payments due to all aspects of credit policy. Then,

$$(A7) \quad 1 - s = \frac{\delta(1-\sigma)(1-\phi)\gamma + \sum_{k=1}^{m} \{X_3/\left[(1+r_1)^{-k}\right]\}}{\sum_{k=1}^{m} \{X_1/\left[(1+r_1)^{-k}\right]\}}.$$

Calculations similar to those used in deriving (A6) yield

$$(A8) \quad X_3 = X_1[1 - s - \delta(1-\sigma)(1-\phi)\gamma].$$

Other subsidies ($s^O = s - s_G$) are modeled as further reductions in annual loan payments. Substitution yields

$$(A9) \quad X_2 = X_3 + \frac{s^O}{\phi \sum_{k=1}^{m} \frac{1}{(1+\rho)^k}}.$$

The effective interest rate for borrowers, r^*, solves

$$(A10) \quad X_3 = \frac{1 - \delta(1-\sigma)(1-\phi)\gamma}{\sum_{k=1}^{m} \left[(1+r^*)^{-k}\right]}.$$

Substitution of (A10) into (A9) and rearranging yields

$$(A11) \quad \frac{1 - [1-(1-\delta)(1-\sigma)](1-\phi)\gamma - s^O}{\phi[1-\delta(1-\sigma)(1-\phi)\gamma]} \times \sum_{k=1}^{m} \frac{1}{(1+r^*)^k} = \sum_{k=1}^{m} \frac{1}{(1+\rho)^k}.$$

Equation (A11) implicitly defines r^*. In the absence of subsidies, (A11) reduces to (A3).

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Government Borrowing

For a guarantee, the government receives $(1-\sigma)(1-\phi)\gamma$ in guarantee fees but expects to pay $(1-\phi)\gamma$ in defaults. Thus, the government cost of a guarantee is s_G as defined by (A6). In addition, when providing the other subsidies described above, the government pays $X_2 - X_3$ with probability ϕ for each period of the loan. The cost to government in each period is $\phi(X_2 - X_3) = \phi s^O X_1 = \phi(s - s_G)X_1$. Total government borrowing requirements, discounted at ρ, are

$$(A12) \quad G_{LG} = s_G + (s - s_G)\phi X_1 \times \sum_{k=1}^{m} \frac{1}{(1+\rho)^k} = s.$$

Under a direct loan program, receipts are X_3 per period with probability ϕ, and 0 with probability $1-\phi$. From (A8) and (A1), $X_3 = X_2 - s^O X_1$; but since $X_2 = X_1$ for a direct loan, $X_3 = (1-s^O)X_1$. Therefore, expected government borrowing is given by

$$(A13) \quad G_{DL} = 1 - \phi(1-s^O) \sum_{k=1}^{m} \frac{X_1}{(1+\rho)^k} = s^O = s.$$

In a tax-exempt program, the government forgoes $X_2 - X_3$ in receipts each period with probability ϕ. Government financing costs are given by

$$(A14) \quad G_{TE} = \phi \sum_{k=1}^{m} \frac{X_2 - X_3}{(1+\rho)^k} = s^O = s.$$

A more detailed version of this Appendix is available upon request.

Appendix B: Parameter Specification

Unless otherwise specified, data are collected from annual versions of "Special Analysis F" of *Special Analyses: Budget of the U.S. Government*. Primary mortgage guarantees issued by the Federal Housing Administration (FHA) and the Veteran's Administration (VA) cover 100 percent of the outstanding principal and interest ($\gamma = 1$) and are paid for by the borrower ($\delta = 1$). Default rates of 3.0 percent were assumed for both programs, based on evidence from the mid-1980's. FHA fees are 3.8 percent; VA fees are 1.9 percent. Weighting by new credit issued yields the value for σ. The overall reduction in borrower payments is estimated to be 1 percent for FHA and 4.2 percent for VA guarantees. Weighting by new credit issued yields an average reduction of 2 percent.

Student loans are 100-percent guaranteed. Loss rates due to default are approximately 9 percent (Bosworth et al., 1987 pp. 134–5). Therefore, $\phi = 0.91$. Students currently pay the full fee of 5.5 percent, implying $\delta = 1$ and $\sigma = 0.39$ $(= 1 - 0.055/0.09)$. Other characteristics of the loan program, particularly the deferral of principal and interest at no cost until graduation, substantially reduce borrower costs. Students would have had to pay 41 percent, 47 percent, and 52 percent more (in present-value terms) for loans in 1984, 1985, and 1986, if there were no guarantee program (table F-12 in *Special Analyses: Budget of the U.S. Government*). Therefore, the guarantee reduced borrower payments (in present value) by 29 percent, 32 percent, and 34 percent respectively, in those years. The value of 32 percent is chosen to represent s.

Small Business Administration (SBA) loan guarantees provide timely payment of 90 percent of outstanding principal and interest. The loss rate on SBA guarantees has been estimated at 9.7 percent (Bosworth et al., 1987 p. 93) and 12.5 percent (Michael Boskin and Bradford Barham, 1984 p. 28). Since fees in the mid-1980's were 1 percent of loan volume and are paid by the bank ($\delta = 0$), a loss rate of 11 percent ($\phi = 0.89$) implies that $\sigma = 0.91$. During 1984–1986, SBA guarantees reduced the present value of borrower payments by 10.8 percent, 14.1 percent, and 18.6 percent (based on data in table F-12). Therefore, s is set equal to 0.14.

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For tax-exempt borrowing, the only parameter to set is s, the overall reduction in borrower payments. If the non-tax-exempt borrowing rate is 10 percent (the value of ρ below), and tax-exempt rates are 7 percent (a spread consistent with results reported in James Poterba [1986]), tax-exempt status reduces the discounted (at 10 percent) value of borrower payments on a 10-year loan by 19 percent. The rate of default for tax-exempts is set equal to zero.

Direct farm lending is meant to represent an amalgam of government lending programs to the agricultural sector. These include programs in the Farmers' Home Administration (FmHA), Rural Electrical Administration, and Export Credits. During 1984–1986, the loan programs reduced borrower payments by 21.4 percent, 20.0 percent, and 30.0 percent (table F-11). A rough average is $s = 0.25$. Calomiris et al. (1986 table A-2) report delinquency rates of approximately 4.0 percent of loan volume for all agricultural credit during 1982–1985. However, federal credit programs in agriculture undoubtedly have higher default rates; the FmHA is estimated to have held 4.5 percent of its portfolio in delinquent loans in 1980, rising to 23.0 percent in 1985 (General Accounting Office, 1986 p. 53). A loss rate of 0.12 ($\phi = 0.88$) is used in the simulation.

The demand elasticity for general borrowers is assumed to be -0.8, based on estimates by Robert Hall (1977), Friedman (1978), and Feldstein and Joosung Jun (1987). In the absence of better data, this value is also used for small businesses. The elasticity of mortgage borrowing is set at -1.8, based on Phoebus Dhrymes and Paul Taubman (1969). Farm-investment elasticity is set at -1.0, based on Michael LeBlanc and James Hrubovcak (1986). The elasticity of student loan demand is estimated from figures in Bosworth et al. (1987). The tax-exempt demand elasticity is based on Patric Hendershott and Timothy Koch (1977). Loan maturities were set at 30 years for mortgages, 10 years for students, small businesses, and tax-exempts, and 20 years for farmers, based on data in table F-11 for 1985 and 1986.

Credit allocations are based on flow-of-funds figures from recent years. Since these data are based on net borrowing, while credit subsidies accompany new credit, some adjustments are necessary. Tax-exempt borrowing represented 8.3 percent of net borrowing during 1980–1987 (see Table 1). FHA and VA mortgages accounted for 4.6 percent of net borrowing since 1980. Agricultural lending represented less than 0.1 percent of net borrowing during 1980–1987, because net agricultural lending was negative in several years. Nonetheless, many new agricultural loans were made. During 1970–1985 and 1976–1980, farm credit averaged 4.1 percent and 4.2 percent of net credit, respectively. Since the government has typically provided between one-third and one-half of all agricultural lending (Council of Economic Advisers, 1986 p. 197), federal farm credit is set at 2 percent of aggregate credit. Net guaranteed student loans for 1980–1987 totalled $29.4 billion, or 0.6 percent of total net borrowing. Net SBA loan guarantees have been negative in recent years, but new SBA disbursements were approximately one-third of guaranteed student lending during 1980–1987, so the SBA allocation is set at 0.2 percent.

Finally, ρ is set at 0.1. Other rates are calculated given equation (A11). Given these parameters, the model may be solved backwards to generate values of a and b_i, the constants in the supply and demand equations. Alternative initial levels of ρ have no appreciable effects on the results, because a and b_i adjust accordingly.

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